

Xiaoyu Yuan

Embodied AI, Neural Decoding, and Multi-Agent Systems
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EDUCATION

University of Oulu *PhD in Computer Science* Mar. 2026 – Present *
Supervisor: Prof. Haoyu Chen Affiliation: Center for Machine Vision and Signal Analysis (CMVS).
University of Helsinki *Master in CS (Algorithms & ML)* GPA: 4/5 (Full Scholarship) 2023 – 2025
Courses: GenAI Lab Project, Neural Networks & Deep Learning, Statistics for Data Science, Design & Analysis of Algorithms, Machine Learning in Molecular Biology, Data Analysis with Python
University of Oulu *Bachelor in Software Engineering* GPA: 4/5 (Double degree) 2019 - 2023
Courses: AI & Software Engineering, Data Modeling & Design, Software Quality & Testing, RASA ChatBot project

PUBLICATIONS

- **X. Yuan**, N. Heikkala, T. Törmänen, H. Järvenoja, G. Zhao, & H. Chen. (2026). MOTOR-Bench: A Real-world Dataset and Multi-agent Framework for Zero-shot Human Mental State Understanding. *In Proc. of the IEEE/CVF Conf. on Computer Vision and Pattern Recognition Workshops (CVPRW 2026)*, pp. 9431–9439. (Oral Presentation)
- **X. Yuan**, Z. Zhang, Y. Sun, Z. Xue, X. Shao, & X. Huang. (2023). A new database of Houma Alliance Book ancient handwritten characters and its baseline algorithm. In Proc. of the 8th Int. Conf. on Multimedia Systems and Signal Processing (ICMSSP '23). ACM. DOI: [10.1145/3613917.3613923](https://doi.org/10.1145/3613917.3613923)
- Z. Zhang, X. Huang, **X. Yuan**, & Y. Sun. (2023). HABFD: Houma Alliance Book facsimiles database. In Proc. of the IEEE Int. Conf. on Image, Vision and Computing (ICIVC '23). IEEE. DOI: [58118.2023.10269984](https://doi.org/58118.2023.10269984)

PATENTS & COPYRIGHTS

- **X. Yuan**, Z. Zhang, X. Huang. (2023). Method for ancient Chinese character sample collection, detection and recognition based on optimized deep network. **Chinese Patent** [CN202311008755.1](https://patent.cnipa.gov.cn/patent/CN202311008755.1) (Published).
- **X. Yuan**, Z. Zhang, X. Huang, et al. (2022). Houma Alliance Book Intelligent Recognition Platform v1.0. **Software Copyright Registration** No. 2022SR1030217, National Copyright Administration of China.

AWARDS

- **1st Prize**, Chinese Collegiate Computing Competition (National-Level, China), 2023
- **2nd Prize**, Jiangsu Province Bachelor Thesis Award (Provincial-Level, China), 2023
- **3rd Prize**, Jiangsu Province University Mathematical Modeling Competition, 2021
- **1st Prize** in the 1st “Prospective Cup” Meta-Intelligent Data Challenge (Track: Entity Recognition in Sedimentology Knowledge Map), held at the CENet2022 Conference, Haikou, China (Nov. 2022).

WORK EXPERIENCE

Human-AI RLHF Evaluation Tasker | Outlier.ai *Freelance, Jan. 2025 – Present*
• Conducted large-scale preference **evaluation of LLM outputs** across tasks (e.g., summarization, Q&A, coding).
• Specialized in **detecting hallucination, evaluating reward models**, and prompting refinement.
• Contribute to **RLHF** and comparison alignment pipelines, focusing on Reward Modeling and cue-response pairing.
Software Tester Intern | Tongdahai Information Technology Co., Ltd. *Nanjing, Jun. 2021 – Jul. 2021*
• Conducted **black-box testing** and analyzed 2000+ **UX feedbacks**
• Collaborated in **Agile sprints** and resolved bugs across platforms

RESEARCH EXPERIENCE

Research Assistant (Project-based) *Remote, University of Oulu*
Project: Visual Theory of Mind in Collaborative Learning with Causal Multi-Agent *Oct. 2025 – Present*
• **Developed SRL-Agents**, a causal multi-agent framework to infer hidden cognitive/emotional states, overcoming the "random guess" performance (20-26% F1) of SOTA MLLMs like InternVL and Gemini.
• **Designed a hierarchical causal architecture** (Behavior→Cognition→Emotion) encoding SRL priors to enforce theory-of-mind reasoning and prevent cue over-interpretation.
• **Achieved 42.77% average F1 score** (77% overall improvement) using InternVL-3.5, with a 107% gain in cognition-specific tasks (19.21% → 39.81%).
• **Managed large-scale inference on CSC (Puhti/LUMI)** high-performance clusters, optimizing V100/A100 GPU usage to process multimodal GroupSRL datasets (video, audio, transcripts).
• **Demonstrated a 17.2% lead** over end-to-end black-box models, validating that theory-driven agent design is superior for abstract reasoning in Embodied AI.
Multi-Task Self-Supervised SLAM for Gastrointestinal Endoscopy Navigation *Helsinki, Finland*
Master Thesis, University of Helsinki *Supervisor: Prof. Laura Ruotsalainen* *Dec. 2024 – Jun. 2025*
• **Reconstructed and adapted a self-supervised learning framework** for simultaneous depth and pose estimation in gastrointestinal endoscopy, addressing challenges of incomplete open-source implementations.
• Implemented a **multi-task architecture** with **ResNet-18 encoder** and **dual attention-based decoder**

branches, jointly optimizing photometric reconstruction, depth smoothness, and **3D pose consistency**.

- **Integrated** brightness-aware masking and spatial attention mechanisms to handle GI-specific challenges: specular reflections, low-texture surfaces (causing feature-based methods to fail), and tissue deformations.
- Achieved a **76% reduction in Absolute Trajectory Error**, demonstrating robust **3D motion tracking** capability in complex environments.
- Experience with remote **GPU** and **HPC** services at the University of Helsinki.

Ancient Character Recognition with Multi-Model Ensemble & Data Augmentation Nanjing, China
Bachelor Thesis, Nanjing Institute of Technology Supervisor: Prof. Xiaohua Huang Jun. 2021 – Jun. 2023
Funded by National College Student Innovation Program, ¥20,000 CNY (~\$2,400 EUR)

- Established **first digital database** of Houma Alliance Book ancient characters (297 classes, 3,547 samples) through collaboration with museum experts and paleographers, demonstrating human-centered approach to addressing data scarcity in historical document digitization.
- **Developed decision-level ensemble framework** fusing deep architectures (ResNet-18, Vision Transformer, Cross-attention Transformer) with linear weight optimization, achieving **94.2%** accuracy on imbalanced dataset.
- Implemented **Mixup-based long-tail optimization** to handle class **imbalance** (head 20% classes contain 60% samples), improving tail-class recognition by **12%** over single-model baselines.
- **Deployed full-stack OCR system** [\[YouTube\]](#) & [\[Github\]](#) with Flask backend and JavaScript frontend, enabling real-time upload, detection, and recognition visualization for historical document researchers.
- **First inventor** on **granted patent**, conference paper in ACM ICMSSP'23 and **software copyright** registration.
- **1st Prize** in Chinese Collegiate Computing Competition, **2nd Prize** in Jiangsu Province Bachelor Thesis Award.

PROJECTS EXPERIENCE

Theoretical Analysis: Aligning 3D Gesture Generation via Preference Optimization 2025
Self-directed Research Project [\[Google Slides\]](#)

- **Conducted a comprehensive study** on applying Direct Preference Optimization (DPO) to **3D gesture** and sign language synthesis to overcome the limitations of traditional reward-based RLHF.
- **Proposed a theoretical framework** to bridge the gap between human evaluation (e.g., GENE Challenge's Mismatching methodology) and model optimization using DPO-style learning.
- **Analyzed the mathematical alignment** between human internal intent and non-verbal behaviors, exploring how to optimize expressiveness and **naturalness** in **3D motion sequences**.
- **Synthesized insights** from SOTA literature to evaluate the trade-offs of using DPO for **high-dimensional generative tasks** in Embodied AI.

AI+BII: Multimodal Generative AI for Architectural Heritage Restoration 2024
Lab team Project [\[YouTube Demo\]](#) & [\[GitHub\]](#)

- Designed **Stable Diffusion-based inpainting system** using VAE-UNet latent diffusion pipeline.
- Implemented **CLIP-guided multimodal conditioning**, achieving high-fidelity text-to-image synthesis and spatial reconstruction.
- Deployed interactive demo on Hugging Face; demonstrated expertise in **GenAI** relevant to motion synthesis.

Multimodal Delphi for EHR Temporal Modeling 2024
Self-study Project [\[Github\]](#)

- Extended Delphi transformer architecture to support **multimodal learning** by integrating 25-dimensional continuous clinical features, enabling unified patient trajectory prediction on Synthea **EHR** cohort.
- Discovered two-stage learning dynamics, suggesting **hierarchical learning in multimodal medical AI**.
- Achieved strong **generalization performance** and demonstrated **effective feature fusion** (continuous features contribute logits difference=4.32), validated through comprehensive model diagnostics.
- Developed complete ML pipeline from data processing (Synthea synthetic EHR) to model training to medical-grade visualization, producing 6-panel analysis figures for **interpretability study**.
- Conducted **interpretability analysis** through **feature contribution attribution**, revealing how continuous clinical features interact with temporal patterns to influence predictions, enabling clinicians to validate.

CancerTypeNet: Tumor Type Classification from Mutational Signatures 2024
Course team project, Machine Learning in Molecular Biology [\[Github\]](#)

- Developed multi-model classification pipeline (DNN, CNN, RF, GB) to predict tumor types from trinucleotide-based **SBS mutational signatures**, achieving **79.5% accuracy** across 22 cancer classes in **PCAWG dataset**.
- Performed dimension reduction and motif-level feature selection to extract sequence-specific mutational patterns.
- Integrated **explainable AI techniques (SHAP, attention heatmaps)** to provide interpretable insights into context-dependent mutational hotspots, enabling oncologists to validate model predictions.
- Demonstrated translational potential for precision oncology applications.

TECHNICAL SKILLS

Generative AI: Latent Diffusion Models (LDM), ControlNet, DPO, RLHF, Multi-Agent Systems, Causal Inference
Multimodal & Neural Decoding: 3D Pose Estimation, Transformer, Signal Processing (EEG/EMG focus)
HPC: CSC (LUMI, Puhti, Mahti), Slurm, Remote GPU Cluster Management, Linux (Bash), Docker.
Deep Learning & Tools: PyTorch, TensorFlow, Hugging Face, OpenMMLab, OpenCV.
Engineering & Systems: Python, C++, Git, FastAPI, Full-stack Prototyping (React/TypeScript)